

Notes on Identifiability

For basis-only measurements, *none of the 6 phi are truly 'dark'* (dark: changing that value of phi produces no change in power measurements). Instead, we obtain:

- 3 equations
- for 6 values of phi

All of the theta values can be uniquely determined. Some experiments and their results:

EXPERIMENT #1: Because we have 3 equations, solve for 3 phi values and freeze them at that. The remaining 3 are randomly chosen such that the 3 equations are satisfied. *Hypothesis: For all randomly chosen 'valid' values of phi, the power measurements should be the same.*

```
regime    : power_basis_only
obs_dim   : 16
free phi   : phi_Ga23, phi_Ga12, phi_Ga34
learn phi  : phi_Gb23, phi_Gb12, phi_Gb34
```

Then, we use our trained model to predict 6 valid values of phi, and we get a starting point:

```
phi=[-1.6254762411117554,
-1.5180786848068237,
-3.113555431365967,
-0.22261983156204224,
0.496172159910202,
1.163313627243042]
```

We now freeze the first 3 values of phi, and find random solutions to the other such that our 3 equations are satisfied.

Our hypothesis is thus true if all of these trials give a similar RMSE (difference between predicted and given powers). We observe:

```
start 0: obs_RMSE=2.447e-03  phi=[-1.6254762411117554, -1.5180786848068237,
-3.113555431365967,          -0.1898368000984192,          0.5241486430168152,
1.5129189491271973]
```

```
start 1: obs_RMSE=4.641e-03  phi=[-1.6254762411117554, -1.5180786848068237,
-3.113555431365967,          -0.1635320782661438,          0.486372172832489,
1.2271262407302856]
```

start 2: obs_RMSE=6.072e-03 phi=[-1.6254762411117554, -1.5180786848068237,
-3.113555431365967, -0.35923558473587036, 0.38156288862228394,
1.0305793285369873]

start 4: obs_RMSE=4.486e-03 phi=[-1.6254762411117554, -1.5180786848068237,
-3.113555431365967, -0.18380194902420044, 0.45512884855270386,
1.2074456214904785]

start 5: obs_RMSE=3.228e-03 phi=[-1.6254762411117554, -1.5180786848068237,
-3.113555431365967, -0.5459941029548645, 0.11541479825973511,
0.8392267227172852]

best obs_RMSE : 2.447e-03

Hence, our hypothesis is true. **What can we do with this?** We can now solve exactly for 3 of the phi values and derive a gauge condition for the other 3. The remaining phi values can be varied freely as long as they obey the gauge condition.

EXPERIMENT #2: We can test for all different kinds of input probes to see if any of the parameters are 'dark'. This is done using the Jacobian matrix, where each column corresponds to one parameter. If any column is all 0, that parameter is 'dark'

For example: for only I4 ([0,0,0,1]) measurements, we find that phi_Ga23, phi_Ga12, phi_Gb12 are completely dark. So, we test random values of these and see if the RMSE changes.

Again, we infer our trained model to get a baseline solution:

*phi_Ga23=1.632441,
*phi_Ga12=1.666195,
phi_Ga34=0.493712,
phi_Gb23=-1.076595,
*phi_Gb12=-2.993508,
phi_Gb34=-0.155018
(*:dark)

Next, we randomly vary the dark phi values, and calculate **max|Δobs|**, the maximum observed difference between the power values predicted by the model using baseline phi values, and the power values obtained by randomly messing with the dark phi values. We observe:

trial 0: max|Δobs|=0.000e+00 phi_dark=[1.860526204109192, 3.245656728744507, 1.5812938213348389]

trial 1: max|Δobs|=2.235e-08 phi_dark=[4.326330184936523, 0.46478271484375, 5.444518089294434]

trial 2: max|Δobs|=0.000e+00 phi_dark=[0.858156681060791, 0.6438947916030884, 1.1564608812332153]

trial 3: max|Δobs|=0.000e+00 phi_dark=[4.564399719238281, 1.9807988405227661, 4.317218780517578]

trial 4: max|Δobs|=0.000e+00 phi_dark=[0.47523069381713867, 1.2355140447616577, 1.9880751371383667]

Since all are close to 0, we can conclude changing these values of phi does nothing to our observed powers. **What can we do with this?** For some specific kind of inputs, we can calculate which values of phi make no difference. We can then not bother with those when programming our PIC.

EXPERIMENT #3: We can also use the 3 equations for phi to generate an 'equivalence class' of phi values. Essentially, if we have a constraint like:

$\Phi_1 + 3 \Phi_2 = 0$, where Φ_1 and Φ_2 are a valid set of phi values predicted, then:

$10\Phi_1$ and $-3.333\Phi_2$ is also a valid solution. We call $\Phi_1 + 3 \Phi_2$ as a 'null' vector of the phi-space, and any *valid* changes along the null dimension should produce similar results. To test this, we derive this vector of phi values:

null #0: +1.000·phi_Ga34 +0.901·phi_Gb23 +0.570·phi_Gb12 +0.430·phi_Ga23
+0.331·phi_Ga12 -0.099·phi_Gb34
** dominant φ direction: phi_Ga34

null #1: +1.000·phi_Gb34 +0.585·phi_Gb23 -0.560·phi_Ga23 +0.440·phi_Ga12
-0.415·phi_Ga34 +0.145·phi_Gb12
** dominant φ direction: phi_Gb34

null #2: -1.000·phi_Ga12 +0.995·phi_Gb12 -0.867·phi_Ga23 -0.133·phi_Gb34
+0.128·phi_Ga34
** dominant φ direction: phi_Ga12

Then, we probe one of those by some value ε ($=0.15$) in the test below, and observe. We see:

Numerical probe along null #0 ($\varepsilon=0.15$): $\max|\Delta_{\text{obs}}| = 2.384\text{e-}07$

Hence, by changing and playing around the null dimension, there is negligible change in the observed powers. **What can we do with this?** Same as above, we can make a certain set of changes to ϕ values and still get the same observable.

Why is this important? For the specific case (such as only one input probe, etc.), we can analytically calculate a gauge condition. But, for multiple probes, it becomes difficult. The model allows us to figure out what freedom we have to make changes to predicted values of ϕ which will still not change the measurement.